Monoalphabetic Substitution Cipher Cryptanalysis

Cryptanalysis Using Frequency, Word Pattern, and Iterative Substitution Analysis (C++)

Course: Cryptography Laboratory (22CPP307)
Instructor: Dr. Meenakshi Tripathi
Group: 9 | Language: C++ (C++17)

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1. Purpose & Experimental Setup

The objective of Assignment 5 is to implement a Monoalphabetic Substitution Cipher and its cryptanalysis using frequency analysis, word patterns, repeated words, and iterative hypothesis testing in C++. As assigned for Group 9, the plaintext was chosen from Page 39 (Group 9 + 30) of 'Introduction to Modern Cryptography' by Jonathan Katz and Yehuda Lindell.

Text Preprocessing and Cleaning Rationale:
The raw textbook page contained non-ASCII characters and LaTeX math symbols (such as Greek Pi, epsilon, greater-than-or-equal, element-of, and prime notation). In classical monoalphabetic substitution ciphers, non-alphabetic characters bypass the 26-letter substitution permutation. Unencrypted Greek and mathematical symbols would cause information leakage and corrupt character frequency distributions. Therefore, all mathematical notations were converted into clean, standard English equivalents (e.g., 'Pi', 'epsilon', 'greater than or equal to'). This ensures authentic, meaningful natural language cryptanalysis without statistical skew or encoding artifacts.

2. Implemented Program Modules

The assignment requires six specific functions/modules, all implemented in modular C++:

- frequency_analysis(): Computes letter counts, descending order frequencies, and percentages.
- word_frequency_analysis(): Extracts 1-letter, 2-letter, 3-letter, and repeated words.
- pattern_analysis(): Identifies letter patterns (e.g., HELLO -> ABCCD) and doubled letters.
- apply_substitution(): Applies candidate mappings, marking unmapped characters with '?'.
- display_partial_plaintext(): Shows ciphertext and partial plaintext side-by-side with mapping table.
- verify_solution(): Re-encrypts plaintext with recovered key and checks for exact ciphertext match.

3. Cryptanalytic Decision Table (Lab Notebook)

The table below documents the systematic, iterative cryptanalytic process used to recover the 26-letter substitution key from the ciphertext (Total: 2,202 letters analyzed):

Step	Observation	Possible Sub	Tested	Result	Decision
1	T occurs most (317, 14.4%)	T -> E	T -> E	Word endings match English patterns	Good
2	ZIT is top 3-letter word (27x)	Z -> T, I -> H	Z->T, I->H	ZIT becomes THE, Z is 2nd freq (9.8%)	Good
3	Q is top 1-letter word (25x)	Q -> A	Q -> A	Q becomes single-letter word A	Good
4	QFR occurs 10x with Q=A	F -> N, R -> D	F->N, R->D	QFR becomes AND; F is 6.0% (N)	Good
5	OL occurs 10x as 2-letter word	O -> I, L -> S	O->I, L->S	OL becomes IS; OZ becomes IT	Good

6	GFT ends in NE (from F, T)	G -> O	G -> O	GFT becomes ONE; GY becomes OF	Good
7	YGK has pattern _ O _	Y -> F, K -> R	Y->F, K->R	YGK becomes FOR; GY becomes OF	Good
8	VIOEI has pattern V H I E H	V -> W, E -> C	V->W, E->C	VIOEI becomes WHICH	Good
9	Title HTKYTEZSN LTEKTZ...	H -> P, N -> Y	H->P, N->Y	Resolves PERFECTLY SECRET ENCRYPTION	Good
10	DTLLQUT has D E S S A _ E	D -> M, U -> G	D->M, U->G	Resolves word MESSAGE	Good
11	WT maps to _ E after 'AND'	W -> B	W -> B	Resolves phrase 'KEY BE A'	Good
12	XFOYGKD has _ N I F O R M	X -> U	X -> U	Resolves word UNIFORM	Good
13	COUTFTKT has _ I G E N E R E	C -> V	C -> V	Resolves cipher name VIGENERE	Good
14	ATN has _ E Y; GIGWDLEG...	A->K, M->Z, B->X	A->K, M->Z	Resolves KEY, RANDOMIZED, EXERCISE	Good

4. Key Recovery & Verification

Recovered Key: QWERTYUIOPASDFGHJKLZXCVBNM

Original Key: QWERTYUIOPASDFGHJKLZXCVBNM

Verification: Re-encrypting recovered plaintext with recovered key matches ciphertext exactly.

Status: VERIFICATION SUCCESSFUL (100% key and text match).

5. Observations

- Frequency analysis provides an excellent initial hypothesis (e.g., T -> E, Z -> T, Q -> A).
- One-letter words ('Q' -> 'A') and three-letter words ('ZIT' -> 'THE') are the most powerful structural anchors.
- Word pattern analysis confirms hypotheses: words like 'VIOEI' (ABACD -> WHICH) and 'DTLLQUT' (ABCCDEB -> MESSAGE) immediately reveal multiple letters.
- Statistical skew from specialized domain vocabulary is overcome by contextual and pattern analysis.

6. Conclusion

Although the Monoalphabetic Substitution Cipher provides a nominal key space of 26! (~4 x 10^26), it offers zero security against cryptanalysis. Because the mapping is 1-to-1 and deterministic, it preserves language frequencies and grammatical word structures. Automated frequency analysis combined with pattern recognition breaks the cipher quickly without brute force. Modern cryptographic systems must introduce confusion and diffusion (as formulated by Claude Shannon) to completely eliminate statistical leakage.